

Picard Iteration for the Characteristic Initial Value Problem in Einstein Equations

Shengrong Wu

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Abstract

We present a first-order Picard iteration for vacuum and Einstein–scalar-field equations in double-null gauge. The numerical realization combines characteristic constraint solves, LGL spectral elements, pole-free spherical operators, Galerkin projection, and independent geometric residual audits.

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1 Introduction

2 Equations and Theoretical Analysis

2.1 Double null foliation and equations

In this section, we introduce the equations for Lorentzian metric under double null foliation. Let $(\mathcal{M}, g)$ be a 4-dimensional Lorentzian manifold. With double null coordinates (u, v, θ^A) , the metric can be written as

$$g = -2\Omega^2(du \otimes dv + dv \otimes du) + g_{AB}(d\theta^A - b^A du) \otimes (d\theta^B - b^B du), \quad (2.1)$$

and we use (g_{AB}, b^A, Ω) to represent the metric. The level sets of u and v are denoted by H_u and $\underline{H}_v$ respectively, and their spherical intersection $H_u \cap \underline{H}_v$ is denoted by $S_{u,v}$. With double null frame

$$e_3 = \Omega^{-1}(\partial_u + b), \quad e_4 = \Omega^{-1}\partial_v, \quad e_A = \partial_{\theta^A},$$

we can define Ricci coefficients, or connection components

$$\begin{aligned} \chi_{AB} &= g(D_A e_4, e_B), & \underline{\chi}_{AB} &= g(D_A e_3, e_B), \\ \eta_A &= -\frac{1}{2}g(D_3 e_A, e_4), & \underline{\eta}_A &= -\frac{1}{2}g(D_4 e_A, e_3), \\ \omega &= -\frac{1}{4}g(D_4 e_3, e_4), & \underline{\omega} &= -\frac{1}{4}g(D_3 e_4, e_3), \\ \zeta_A &= \frac{1}{2}g(D_A e_4, e_3) = -\frac{1}{4}\Omega^{-2}(\mathcal{L}_v b^B)g_{AB}. \end{aligned} \tag{2.2}$$

Their transport equations are listed below:

$$\begin{aligned} \nabla_4 \operatorname{tr} \chi + \frac{1}{2}(\operatorname{tr} \chi)^2 &= -\operatorname{Ric}_{44} - |\hat{\chi}|^2 - 2\omega \operatorname{tr} \chi, \\ \nabla_3 \operatorname{tr} \underline{\chi} + \frac{1}{2}(\operatorname{tr} \underline{\chi})^2 &= -\operatorname{Ric}_{33} - |\hat{\underline{\chi}}|^2 - 2\underline{\omega} \operatorname{tr} \underline{\chi}, \end{aligned} \tag{2.3}$$

$$\begin{aligned} \nabla_3 \hat{\chi}_{AB} + \frac{1}{2} \operatorname{tr} \underline{\chi} \hat{\chi}_{AB} &= \widehat{\operatorname{Ric}}_{AB} + 2\underline{\omega} \hat{\chi}_{AB} + (\nabla \hat{\otimes} \eta)_{AB} + (\eta \hat{\otimes} \eta)_{AB} - \frac{1}{2} \operatorname{tr} \chi \hat{\chi}_{AB}, \\ \nabla_4 \hat{\underline{\chi}}_{AB} + \frac{1}{2} \operatorname{tr} \chi \hat{\underline{\chi}}_{AB} &= \widehat{\operatorname{Ric}}_{AB} + 2\omega \hat{\underline{\chi}}_{AB} + (\nabla \hat{\otimes} \underline{\eta})_{AB} + (\underline{\eta} \hat{\otimes} \underline{\eta})_{AB} - \frac{1}{2} \operatorname{tr} \underline{\chi} \hat{\underline{\chi}}_{AB}, \end{aligned} \tag{2.4}$$

$$\begin{aligned} \nabla_3 \operatorname{tr} \chi &= -\frac{1}{2} \operatorname{tr} \chi \operatorname{tr} \underline{\chi} + \not{g}^{AB} R_{3AB4} + 2\underline{\omega} \operatorname{tr} \chi + 2\hat{\chi} \cdot \underline{\eta} + 2|\eta|^2 - \hat{\chi} \cdot \hat{\underline{\chi}} \\ &= -\operatorname{tr} \chi \operatorname{tr} \underline{\chi} + 2\underline{\omega} \operatorname{tr} \chi + 2\hat{\chi} \cdot \underline{\eta} + 2|\eta|^2 - 2K + R + \operatorname{Ric}_{34}, \\ \nabla_4 \operatorname{tr} \underline{\chi} &= -\frac{1}{2} \operatorname{tr} \chi \operatorname{tr} \underline{\chi} + \not{g}^{AB} R_{3AB4} + 2\omega \operatorname{tr} \underline{\chi} + 2\underline{\hat{\chi}} \cdot \eta + 2|\underline{\eta}|^2 - \hat{\chi} \cdot \hat{\underline{\chi}} \\ &= -\operatorname{tr} \chi \operatorname{tr} \underline{\chi} + 2\omega \operatorname{tr} \underline{\chi} + 2\underline{\hat{\chi}} \cdot \eta + 2|\underline{\eta}|^2 - 2K + R + \operatorname{Ric}_{34}. \end{aligned} \tag{2.5}$$

$$\begin{aligned} \nabla_4 \eta &= -\chi \cdot (\eta - \underline{\eta}) - \frac{1}{2} R_{A434}, \\ \nabla_3 \underline{\eta} &= -\underline{\chi} \cdot (\underline{\eta} - \eta) - \frac{1}{2} R_{A343}, \\ \nabla_4 \underline{\omega} &= \frac{1}{4} (\operatorname{Ric}_{34} + \not{g}^{AB} R_{3AB4}) + 2\underline{\omega} \underline{\omega} + \frac{1}{2} |\eta|^2 - \eta \cdot \underline{\eta} \\ &= 2\underline{\omega} \underline{\omega} - \frac{1}{2} K + \frac{1}{2} \operatorname{Ric}_{34} + \frac{1}{4} R + \frac{1}{4} \hat{\chi} \cdot \hat{\underline{\chi}} - \frac{1}{8} \operatorname{tr} \chi \operatorname{tr} \underline{\chi} + \frac{1}{2} |\eta|^2 - \eta \cdot \underline{\eta}, \\ \nabla_3 \omega &= \frac{1}{4} (\operatorname{Ric}_{34} + \not{g}^{AB} R_{3AB4}) + 2\underline{\omega} \omega + \frac{1}{2} |\underline{\eta}|^2 - \eta \cdot \underline{\eta} \\ &= 2\omega \underline{\omega} - \frac{1}{2} K + \frac{1}{2} \operatorname{Ric}_{34} + \frac{1}{4} R + \frac{1}{4} \hat{\chi} \cdot \hat{\underline{\chi}} - \frac{1}{8} \operatorname{tr} \chi \operatorname{tr} \underline{\chi} + \frac{1}{2} |\underline{\eta}|^2 - \eta \cdot \underline{\eta}. \end{aligned} \tag{2.6}$$

Moreover, we have equations for ζ :

$$\begin{aligned} \Omega \operatorname{Ric}_{3A} &= \Omega \nabla_3 \zeta + \frac{3}{2} \Omega \operatorname{tr} \underline{\chi} \zeta + \Omega \hat{\underline{\chi}} \cdot \zeta + \Omega \operatorname{tr} \underline{\chi} \nabla \log \Omega \\ &\quad + 2\nabla(\Omega \underline{\omega}) + \operatorname{div}(\Omega \hat{\underline{\chi}}) - \frac{1}{2} \nabla(\Omega \operatorname{tr} \underline{\chi}), \\ \Omega \operatorname{Ric}_{4A} &= -\Omega \nabla_4 \zeta - \frac{3}{2} \Omega \operatorname{tr} \chi \zeta - \Omega \hat{\chi} \cdot \zeta + \Omega \operatorname{tr} \chi \nabla \log \Omega \\ &\quad + 2\nabla(\Omega \omega) + \operatorname{div}(\Omega \hat{\chi}) - \frac{1}{2} \nabla(\Omega \operatorname{tr} \chi). \end{aligned} \tag{2.7}$$

Here K stands for Gaussian curvature of topological sphere $S_{u,v}$. Gauss-Codazzi equations imply

$$K = \frac{1}{2} g^{AB} R_{3A4B} + \frac{1}{2} R + \frac{1}{2} \text{Ric}_{34} + \frac{1}{2} \hat{\chi} \cdot \hat{\chi} - \frac{1}{4} \text{tr } \chi \text{tr } \underline{\chi}, \quad (2.8)$$

$$\nabla^B \hat{\chi}_{AB} - \frac{1}{2} \nabla_A \text{tr } \chi = -\frac{1}{2} R_{A434} + \text{Ric}_{4A} + \frac{1}{2} \text{tr } \chi \zeta_A - \zeta^B \hat{\chi}_{AB}, \quad (2.9)$$

$$\begin{aligned} \nabla^B \hat{\chi}_{AB} - \frac{1}{2} \nabla_A \text{tr } \underline{\chi} &= -\frac{1}{2} R_{A343} + \text{Ric}_{3A} - \frac{1}{2} \text{tr } \underline{\chi} \zeta_A + \zeta^B \hat{\chi}_{AB}. \\ \nabla_A \eta_B - \nabla_B \eta_A &= R_{4[AB]3} + \hat{\chi}^C_{[A} \hat{\chi}_{B]C}, \\ \nabla_A \underline{\eta}_B - \nabla_B \underline{\eta}_A &= -R_{4[AB]3} - \hat{\chi}^C_{[A} \hat{\chi}_{B]C}, \end{aligned} \quad (2.10)$$

For Weyl curvature,

$$W_{\mu\nu\theta\lambda} = R_{\mu\nu\theta\lambda} - \frac{1}{2} (\text{Ric}_{\mu\theta} g_{\nu\lambda} + g_{\mu\theta} \text{Ric}_{\nu\lambda} - \text{Ric}_{\nu\theta} g_{\mu\lambda} - g_{\nu\theta} \text{Ric}_{\mu\lambda}) + \frac{R}{6} (g_{\mu\theta} g_{\nu\lambda} - g_{\nu\theta} g_{\mu\lambda}),$$

we define the curvature components

$$\begin{aligned} \beta_A &= \frac{1}{2} W(e_A, e_4, e_3, e_4), \quad \underline{\beta}_A = \frac{1}{2} W(e_A, e_3, e_3, e_4), \\ \rho &= \frac{1}{4} W(e_4, e_3, e_4, e_3), \quad \sigma = \frac{1}{4} {}^* W(e_4, e_3, e_4, e_3). \end{aligned} \quad (2.11)$$

Then previous equations imply the following algebraic relations,

$$K = -\rho + \frac{1}{2} R + \text{Ric}_{34} + \frac{1}{2} \hat{\chi} \cdot \hat{\chi} - \frac{1}{4} \text{tr } \underline{\chi} \text{tr } \chi, \quad (2.12)$$

$$\text{curl } \eta = -\text{curl } \underline{\eta} = \sigma + \frac{1}{2} \hat{\chi} \wedge \hat{\chi}, \quad (2.13)$$

$$\text{div } \hat{\chi} - \frac{1}{2} \nabla \text{tr } \chi = -\beta + \frac{1}{2} \text{Ric}_4 + \frac{1}{2} \text{tr } \chi \zeta - \zeta \cdot \hat{\chi}, \quad (2.14)$$

$$\text{div } \hat{\chi} - \frac{1}{2} \nabla \text{tr } \underline{\chi} = \underline{\beta} + \frac{1}{2} \text{Ric}_3 - \frac{1}{2} \text{tr } \underline{\chi} \zeta + \zeta \cdot \hat{\chi}. \quad (2.15)$$

After introducing the reduced curvature

$$\beta_A^r = \beta_A - \frac{1}{2} \text{Ric}_{4A}, \quad \underline{\beta}_A^r = \underline{\beta}_A + \frac{1}{2} \text{Ric}_{3A}, \quad \sigma^r = \sigma + \frac{1}{2} \hat{\chi} \wedge \hat{\chi}, \quad (2.16)$$

we can write the equations for curvature components

$$\begin{aligned} \Omega \nabla_3 K + \Omega \text{tr } \underline{\chi} K &= \text{div } (\Omega \underline{\beta}^r) + \frac{1}{2} \text{div } (\eta \Omega \text{tr } \underline{\chi}) - \text{div } (\eta \cdot \Omega \hat{\chi}), \\ \Omega \nabla_4 K + \Omega \text{tr } \chi K &= -\text{div } (\Omega \beta^r) + \frac{1}{2} \text{div } (\underline{\eta} \Omega \text{tr } \chi) - \text{div } (\underline{\eta} \cdot \Omega \hat{\chi}), \end{aligned} \quad (2.17)$$

$$\begin{aligned} \nabla_3 \beta^r + (\text{tr } \underline{\chi} - 2\underline{\omega}) \beta^r &= -\nabla K + {}^* \nabla \sigma + 2 \hat{\chi} \cdot \underline{\beta} \\ &\quad + \frac{1}{2} (\nabla(\hat{\chi} \cdot \hat{\chi}) - {}^* \nabla(\hat{\chi} \wedge \hat{\chi})) - \frac{1}{4} \nabla (\text{tr } \chi \text{tr } \underline{\chi}) \\ &\quad + 3(\eta \rho + {}^* \eta \sigma) + \nabla_A (g^{CD} \text{Ric}_{CD}) - \nabla^B \text{Ric}_{BA} - \frac{1}{2} \text{tr } \underline{\chi} \text{Ric}_{4A}, \\ \nabla_4 \underline{\beta}^r + (\text{tr } \chi - 2\omega) \underline{\beta}^r &= \nabla K + {}^* \nabla \sigma + 2 \hat{\chi} \cdot \beta \\ &\quad - \frac{1}{2} (\nabla(\hat{\chi} \cdot \hat{\chi}) + {}^* \nabla(\hat{\chi} \wedge \hat{\chi})) + \frac{1}{4} \nabla (\text{tr } \chi \text{tr } \underline{\chi}) \\ &\quad - 3(\underline{\eta} \rho - {}^* \underline{\eta} \sigma) - \nabla_A (g^{CD} \text{Ric}_{CD}) + \nabla^B \text{Ric}_{BA} + \frac{1}{2} \text{tr } \chi \text{Ric}_{3A}. \end{aligned} \quad (2.18)$$

Using (2.6), we derive

$$\begin{aligned}\Omega \nabla_3 \operatorname{curl} \underline{\eta} + \Omega \operatorname{tr} \underline{\chi} \operatorname{curl} \underline{\eta} &= \operatorname{curl}(\Omega \underline{\beta}^r) + \epsilon^{AC} \nabla_A (\Omega \underline{\chi} \cdot \underline{\eta})_C - \epsilon^{AC} \nabla_A (\Omega \operatorname{Ric})_{C3}, \\ \Omega \nabla_4 \operatorname{curl} \eta + \Omega \operatorname{tr} \chi \operatorname{curl} \eta &= -\operatorname{curl}(\Omega \beta^r) + \epsilon^{AC} \nabla_A (\Omega \chi \cdot \underline{\eta})_C - \epsilon^{AC} \nabla_A (\Omega \operatorname{Ric})_{C4},\end{aligned}\quad (2.19)$$

and

$$\begin{aligned}\Omega \nabla_3 (\operatorname{div} \underline{\eta} - K) + \Omega \operatorname{tr} \underline{\chi} (\operatorname{div} \underline{\eta} - K) &= 4 \operatorname{div}(\Omega \widehat{\chi} \cdot \zeta) - \operatorname{div}(\Omega \operatorname{Ric}_3), \\ \Omega \nabla_4 (\operatorname{div} \eta - K) + \Omega \operatorname{tr} \chi (\operatorname{div} \eta - K) &= -4 \operatorname{div}(\Omega \widehat{\chi} \cdot \zeta) - \operatorname{div}(\Omega \operatorname{Ric}_4).\end{aligned}\quad (2.20)$$

For a scalar function ϕ , there is equation

$$-\Omega^2 \square \phi = \Omega e_3(\Omega e_4 \phi) + \frac{1}{2} \Omega \operatorname{tr} \chi \Omega e_3 \phi + \frac{1}{2} \Omega \operatorname{tr} \underline{\chi} \Omega e_4 \phi - \Omega^2 \Delta \phi - 2\Omega^2 \eta \cdot \nabla \phi. \quad (2.21)$$

2.2 Local existence

We now introduce some results on Einstein vacuum equations. Similar results can be derived for other Einstein field equations and we omit them due to length of the article.

We first state the existence theorem for characteristic initial value problem of Einstein vacuum equations from [3].

Theorem 2.1. *Given initial null hypersurface $H_0, \underline{H}_0$ and regular initial metric (g, Ω, b) on it, we define $\chi_{AB} = \frac{1}{2} \mathcal{L}_{e_4} g_{AB}$ on H_0 and $\underline{\chi}_{AB} = \frac{1}{2} \mathcal{L}_{e_3} g_{AB}$ on $\underline{H}_0$. Suppose*

$$e_4(\operatorname{tr} \chi) + \frac{1}{2}(\operatorname{tr} \chi)^2 + |\widehat{\chi}|^2 + 2\omega \operatorname{tr} \chi = 0, \quad e_3(\operatorname{tr} \underline{\chi}) + \frac{1}{2}(\operatorname{tr} \underline{\chi})^2 + |\widehat{\underline{\chi}}|^2 + 2\underline{\omega} \operatorname{tr} \underline{\chi},$$

hold on $H_0, \underline{H}_0$ respectively, then there is a future open neighborhood of $H_0 \cup \underline{H}_0$ and metric (g, b, Ω) on it solving the Einstein vacuum equations and satisfying the initial data.

This theorem reveals the local existence, and we further remark that for larger region, such as $(0, 1)^2 \times \mathbb{S}^2$, as long as the a priori estimates hold in it, the solution can be extended to the region.

2.3 Residual control

In this paper, we target to deal with asymmetric Einstein equations, which in general do not have explicit solutions. To measure the computational error, we use Ricci residues. For Einstein equation

$$G_{\mu\nu} := \operatorname{Ric}_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = T_{\mu\nu},$$

we measure $\|G - T\|_{L^2(S_{u,v})}$ to evaluate the approximation. The theorems below reveals that in our case the difference of metrics is controlled by the difference of curvature residues.

Theorem 2.2. *We consider metric $\overset{\circ}{g}_{AB}, \overset{\circ}{b}, \overset{\circ}{\Omega}$ on $(0, 1) \times (0, 1) \times \mathbb{S}^2$ such that*

$$\overset{\circ}{\operatorname{Ric}}_{44}(0, v) = 0, \quad \overset{\circ}{\operatorname{Ric}}_{33}(u, 0) = 0, \quad \overset{\circ}{\operatorname{Ric}}_{4A}(0, v) = 0, \quad \overset{\circ}{\operatorname{Ric}}_{3A}(u, 0) = 0,$$

and estimates

$$\|\overset{\circ}{\operatorname{Ric}}_{AB}, \overset{\circ}{\operatorname{Ric}}_3, \overset{\circ}{\operatorname{Ric}}_4, \overset{\circ}{\operatorname{Ric}}_{34}, \overset{\circ}{\operatorname{Ric}}_{33}, \overset{\circ}{\operatorname{Ric}}_{44}\|_{L_u^\infty L_v^\infty H^3(S)} \leq \epsilon. \quad (2.22)$$

We assume moreover

$$\|\partial_{\mathbb{S}} \overset{\circ}{g}, \partial_{\mathbb{S}} \overset{\circ}{b}, \partial_{\mathbb{S}} \overset{\circ}{\Omega}, \psi\|_{L_u^\infty L_v^\infty H^4(S)} \leq C_0, \quad (2.23)$$

where ψ stands for Ricci coefficients in interest. Then for $\epsilon = \epsilon(C_0)$ sufficiently small, there exists (g, b, Ω) solving EVE on $(0, 1) \times (0, 1) \times \mathbb{S}^2$ with initial data

$$(g, b, \Omega) = (\overset{\circ}{g}, \overset{\circ}{b}, \overset{\circ}{\Omega}), \text{ on } \underline{H}_0 \cup H_0,$$

and estimates

$$\|g - \overset{\circ}{g}, b - \overset{\circ}{b}, \Omega - \overset{\circ}{\Omega}\|_{L_u^\infty L_v^\infty H^4(S)} \leq \epsilon^{1/2}. \quad (2.24)$$

The methods of deriving the equations of difference and doing estimates are similar to section 7 of [2], so we only write the proof in sketch.

Define auxiliary functions $\omega^\dagger, \underline{\omega}^\dagger$ by

$$\nabla_3 \omega^\dagger = \frac{1}{2} \text{curl} \zeta, \quad \omega^\dagger|_{H_{-1}} = 0, \quad \nabla_4 \underline{\omega}^\dagger = \frac{1}{2} \text{curl} \zeta, \quad \underline{\omega}^\dagger|_{\underline{H}_0} = 0.$$

We make bootstrap assumption

$$\begin{aligned} & \|\widetilde{\beta}, \widetilde{K}, \nabla \widetilde{\eta}, \nabla \widetilde{\eta}, \nabla \widetilde{\chi}, \nabla \widetilde{\omega}, \nabla \widetilde{\omega}^\dagger\|_{L_v^2 H^3(S)} \\ & + \|\widetilde{K}, \widetilde{\beta}, \nabla \widetilde{\eta}, \nabla \widetilde{\eta}, \nabla \widetilde{\chi}, \nabla \widetilde{\omega}, \nabla \widetilde{\omega}^\dagger\|_{L_u^2 H^3(S)} \leq B \exp(D(v+u))\epsilon. \end{aligned}$$

We will prove that we can choose D sufficiently large so that the bootstrap constant B can be improved to $B \leq 1$.

The motivation is direct. Let $\partial_u f \leq C_0 f, \partial_v f \leq C_0 f$, and suppose

$$f(u, v) \leq B \exp(D(v+u))\epsilon,$$

then we integrate it

$$\begin{aligned} f(u, v) - f(0, v) & \leq C_0 \int_0^u f(u', v) du' \\ & \leq \frac{C_0 B}{D} (\exp(D(u+v)) - \exp(Dv)) \epsilon \leq \frac{C_0 B}{D} \exp(D(u+v)) \epsilon, \\ f(u, v) - f(u, 0) & \leq C_0 \int_0^v f(u, v') dv' \\ & \leq \frac{C_0 B}{D} (\exp(D(u+v)) - \exp(Du)) \epsilon \leq \frac{C_0 B}{D} \exp(D(u+v)) \epsilon. \end{aligned}$$

If we choose $D^{1/2} \geq C_0 B$, the bootstrap assumption will be closed. Absorbing the bootstrap constant by the parameter D actually shows the increasing rate is controlled exponentially. Even though the norms increase quickly, they will not exceed $O(\epsilon) \leq o(\epsilon^{1/2})$ in finite region.

In summary, we conclude

Lemma 2.3. Suppose $\|f\|_{L^2(S_{u,v})} \leq B \exp(D(v+u))\epsilon$, then we have

$$\|f\|_{L_u^2 L^2(S)} + \|f\|_{L_v^2 L^2(S)} \leq D^{-1/2} \exp(D(v+u))\epsilon, \quad (2.25)$$

when D is sufficiently large.

Suppose either $\|h\|_{L_u^2 L^2(S)} \leq B \exp(D(u+v))\epsilon$ or $\|h\|_{L_v^2 L^2(S)} \leq B \exp(D(u+v))\epsilon$, then we have

$$\|h\|_{L_u^2 L_v^2 L^2(S)} \leq D^{-1/2} \exp(D(v+u))\epsilon, \quad (2.26)$$

when D is sufficiently large.

We use notation

$$\psi_1 \in \{\eta, \underline{\eta}, \chi, \omega\}, \quad \psi_2 \in \{\eta, \underline{\eta}, \underline{\chi}, \underline{\omega}\}$$

Because

$$\mathcal{L}_v b = -2\Omega^2(\eta - \underline{\eta}), \quad \mathcal{L}_v g = 2\Omega\chi, \quad \mathcal{L}_v \log \Omega = -2\Omega\omega,$$

we obtain that

$$\|\widetilde{b}, \widetilde{g}, \widetilde{\log \Omega}\|_{H^4(S_{u,v})} \lesssim \|\widetilde{\eta}, \widetilde{\underline{\eta}}, \widetilde{\chi}, \widetilde{\omega}, \widetilde{\Omega}, \widetilde{g}\|_{L_v^2 H^4(S)} \lesssim B \exp(D(u+v))\epsilon. \quad (2.27)$$

For Bianchi pair (Ψ_1, Ψ_2) in interest, we consider equations

$$\begin{aligned} \Omega \nabla_3 \widetilde{\Psi}_1 + \mathcal{D} \widetilde{\Psi}_2 &= \widetilde{\psi} \widetilde{\Psi} + \widetilde{\psi} \nabla \widetilde{\psi} + \widetilde{\psi}^3 + \widetilde{b} \cdot \nabla \widetilde{\Psi}_1 + \widetilde{\partial g} \widetilde{\Psi}_2 + \widetilde{g} \nabla \widetilde{\Psi}_2 + \widetilde{\psi} \overset{\circ}{\text{Ric}} + \nabla \overset{\circ}{\text{Ric}}, \\ \Omega \nabla_4 \widetilde{\Psi}_2 - {}^* \mathcal{D} \widetilde{\Psi}_1 &= \widetilde{\psi} \widetilde{\Psi} + \widetilde{\psi} \nabla \widetilde{\psi} + \widetilde{\psi}^3 + \widetilde{\partial g} \widetilde{\Psi}_1 + \widetilde{g} \nabla \widetilde{\Psi}_1 + \widetilde{\psi} \overset{\circ}{\text{Ric}} + \nabla \overset{\circ}{\text{Ric}}. \end{aligned}$$

Energy estimates give that

$$\begin{aligned} \|\widetilde{\Psi}_1\|_{L_v^2 H^3(S)}^2 + \|\widetilde{\Psi}_2\|_{L_u^2 H^3(S)}^2 \\ \lesssim \epsilon^2 + \|\widetilde{\partial g}, \widetilde{g}, \widetilde{\psi}, \nabla \widetilde{\psi}, \widetilde{\Psi}, \widetilde{b}\|_{L_u^2 L_v^2 H^3(S)}^2 \lesssim (1 + D^{-1/2}) \exp(2D(u+v))\epsilon^2. \end{aligned}$$

Applying the estimate to pairs

$$(\beta, (K, \text{curl } \eta)), \quad ((K, \text{curl } \underline{\eta}), \underline{\beta}), \quad (\nabla \text{tr } \underline{\chi}, 0), \quad (0, \nabla \text{tr } \chi),$$

$$(\text{div } \underline{\eta} - K, 0), \quad (0, \text{div } \eta - K), \quad (\nabla \omega + {}^* \nabla \omega^\dagger - \frac{1}{2} \beta, 0), \quad (0, -\nabla \underline{\omega} + {}^* \nabla \underline{\omega}^\dagger - \frac{1}{2} \underline{\beta}),$$

will finish the proof.

Remark 2.4. The conditions

$$\overset{\circ}{\text{Ric}}_{44}(-1, v) = 0, \quad \overset{\circ}{\text{Ric}}_{33}(u, 0) = 0,$$

guarantee that $(\overset{\circ}{g}, \overset{\circ}{b}, \overset{\circ}{\Omega})$ is a valid initial data set for characteristic initial value problem of Einstein vacuum equation, and

$$\overset{\circ}{\text{Ric}}_{4A}(-1, v) = 0, \quad \overset{\circ}{\text{Ric}}_{3A}(u, 0) = 0,$$

ensure that $\overset{\circ}{\zeta}$ along H_{-1} and $\underline{H}_0$ is equal to the ζ of corresponding EVE solution.

3 Picard Iteration Design and Numerical Strategies

In this section, we introduce the iteration design and numerical strategies.

3.1 Picard iteration for Einstein equations

In this section, we first consider Einstein scalar-field equations

$$\text{Ric} = d\phi \otimes d\phi, \quad \square_g \phi = 0.$$

We work in region

$$Q = \{-1 < u < u_0, \quad 0 < v < v_0(u)\}.$$

The first task is to construct a valid initial data for ESE. We directly prescribe g, b, Ω along $\underline{H}_0$. The only requirement is

$$-\Omega \nabla_3 (\Omega \text{tr } \underline{\chi}) - \frac{1}{2} (\Omega \text{tr } \underline{\chi})^2 - |\Omega \widehat{\underline{\chi}}|^2 - 4\Omega \underline{\omega} \Omega \text{tr } \underline{\chi} \geq 0$$

Then $\Omega e_3 \phi(u, 0)$ can be assigned via equation

$$\Omega e_3 \phi = \left(-\Omega \nabla_3 (\Omega \operatorname{tr} \underline{\chi}) - \frac{1}{2} (\Omega \operatorname{tr} \underline{\chi})^2 - |\Omega \widehat{\underline{\chi}}|^2 - 4\Omega \underline{\omega} \Omega \operatorname{tr} \underline{\chi} \right)^{1/2}.$$

After defining $\phi(-1, 0)$ and $\zeta(-1, 0)$, we derive $\phi(u, 0)$ from $\Omega e_3 \phi(u, 0)$ and $\zeta(u, 0)$ via

$$\Omega \nabla_3 \zeta + \frac{3}{2} \Omega \operatorname{tr} \underline{\chi} \zeta + \Omega \widehat{\underline{\chi}} \cdot \zeta = -2\nabla(\Omega \underline{\omega}) - \operatorname{div}(\Omega \widehat{\underline{\chi}}) + \frac{1}{2} \nabla(\Omega \operatorname{tr} \underline{\chi}) - \Omega \operatorname{tr} \underline{\chi} \nabla \log \Omega + \Omega e_3 \phi \nabla \phi.$$

Thus $\eta = \zeta + \nabla \log \Omega$ and $\underline{\eta} = -\zeta + \nabla \log \Omega$ are also prescribed. We then define $\Omega \operatorname{tr} \chi(-1, 0)$ and $\Omega \widehat{\chi}(-1, 0)$, and compute $\Omega \chi(u, 0)$ by

$$\Omega \nabla_3 (\Omega \operatorname{tr} \chi) + \Omega \operatorname{tr} \underline{\chi} \Omega \operatorname{tr} \chi = \Omega^2 \left(2 \operatorname{div} \eta + 2 |\eta|^2 - 2K(g) + |\nabla \phi|^2 \right),$$

$$\Omega \nabla_3 (\Omega \widehat{\chi}) + \frac{1}{2} \Omega \operatorname{tr} \underline{\chi} \Omega \widehat{\chi} = \Omega^2 (\nabla \hat{\otimes} \eta + \eta \hat{\otimes} \eta) - \frac{1}{2} \Omega \operatorname{tr} \chi \Omega \widehat{\underline{\chi}} + \frac{1}{2} \Omega^2 \nabla \phi \hat{\otimes} \nabla \phi.$$

We also directly assign $\Omega e_4 \phi(-1, 0)$ and $\Omega \omega(-1, 0)$, and compute them along $\underline{H}_0$ via

$$\Omega e_3 (\Omega e_4 \phi) + \frac{1}{2} \Omega \operatorname{tr} \chi \Omega e_3 \phi + \frac{1}{2} \Omega \operatorname{tr} \underline{\chi} \Omega e_4 \phi - \Omega^2 \Delta \phi - 2\Omega^2 \eta \cdot \nabla \phi = 0,$$

$$\frac{1}{2} \Omega e_3 \phi \Omega e_4 \phi + \frac{1}{2} \Omega^2 |\nabla \phi|^2 = 2\Omega \nabla_3 (\Omega \omega) + \Omega^2 K - \frac{1}{2} \Omega \widehat{\chi} \cdot \Omega \widehat{\underline{\chi}} + \frac{1}{4} \Omega \operatorname{tr} \chi \Omega \operatorname{tr} \underline{\chi} - \Omega^2 |\underline{\eta}|^2 + 2\Omega^2 \eta \cdot \underline{\eta}.$$

Along H_{-1} , we prescribe $\Omega e_4 \phi(-1, v)$ and $\Omega \omega(-1, v)$ to compute ϕ, Ω . Consider a symmetric and $g(-1, 0)$ -trace-free tensor $X_{AB}(-1, v)$. We then prescribe $g, \Omega \chi$ along H_{-1} by solving:

$$\begin{cases} \Omega \widehat{\chi}_{AB} = \frac{1}{2} \left(X_{AC} (g(-1, 0))^{CD} g_{DB} + X_{BC} (g(-1, 0))^{CD} g_{DA} \right), \\ \partial_v (\Omega \operatorname{tr} \chi) = -\frac{1}{2} (\Omega \operatorname{tr} \chi)^2 - 4\Omega \omega \Omega \operatorname{tr} \chi - (\Omega e_4 \phi)^2 - \Omega \widehat{\chi}_{AB} \Omega \widehat{\chi}_{CD} g^{AC} g^{BD}, \\ \mathcal{L}_v g_{AB} = \Omega \operatorname{tr} \chi g_{AB} + 2\Omega \widehat{\chi}_{AB}. \end{cases} \quad (3.1)$$

With these initial data, we can start the iteration. The quantities to be iterated are

$$g^{(i)}, (b^{(i)})^A, \log \Omega^{(i)}, (\Omega \chi)_{AB}^{(i)}, (\Omega \underline{\chi})_{AB}^{(i)}, (\zeta^{(i)})^A, (\Omega \omega)^{(i)}, (\Omega \underline{\omega})^{(i)}, (\Omega e_3 \phi)^{(i)}, (\Omega e_4 \phi)^{(i)}, \nabla_A \phi^{(i)}.$$

Initially, we set $g^{(0)} = g|_{v=0}$, $\Omega^{(0)} = \Omega|_{v=0}$, $\zeta^{(0)} = \zeta|_{v=0}$,

$$(b^{(0)})^A(u, v) = b^A(u, 0) - \int_0^v 4(\Omega^{(0)})^2 (\zeta^{(0)})^A(u, v') \, dv',$$

$\widehat{\chi}^{(0)} = \widehat{\chi}|_{v=0}$, $\operatorname{tr} \chi^{(0)} = \operatorname{tr} \chi|_{v=0}$, $\widehat{\underline{\chi}}^{(0)} = \widehat{\underline{\chi}}|_{v=0}$, $\operatorname{tr} \underline{\chi}^{(0)} = \operatorname{tr} \underline{\chi}|_{v=0}$, $\omega^{(0)} = \omega|_{v=0}$, $\underline{\omega}^{(0)} = \underline{\omega}|_{v=0}$, $(e_3 \phi)^{(0)} = e_3 \phi|_{v=0}$, $(e_4 \phi)^{(0)} = e_4 \phi|_{v=0}$, $(\nabla_A \phi)^{(0)} = \nabla_A \phi|_{v=0}$. For all i , we always require

$$\eta_A^{(i)} = (\zeta^{(i)})^B g_{AB}^{(i)} + \nabla_A \log \Omega^{(i)}, \quad \underline{\eta}_A^{(i)} = -(\zeta^{(i)})^B g_{AB}^{(i)} + \nabla_A \log \Omega^{(i)}$$

1. Construction of $(\Omega \widehat{\chi})^{(i+1/2)}$, $\partial_v \phi^{(i+1)}$, and $\nabla_A \phi^{(i+1)}$

We use (2.4) to define $(\Omega \widehat{\chi})^{(i+1/2)}$,

$$\begin{cases} \mathcal{L}_{\partial_u + b^{(i)}} (\Omega \widehat{\chi}^{(i+1/2)})_{AB} - \frac{1}{2} (\Omega \operatorname{tr} \underline{\chi})^{(i)} (\Omega \widehat{\chi}^{(i+1/2)})_{AB} \\ \quad - (\Omega \widehat{\underline{\chi}}^{(i)})_A{}^C (\Omega \widehat{\chi}^{(i+1/2)})_{CB} - (\Omega \widehat{\underline{\chi}}^{(i)})_B{}^C (\Omega \widehat{\chi}^{(i+1/2)})_{CA} \\ \quad = (\Omega^{(i)})^2 \left(\left(\nabla^{(i)} \hat{\otimes} \eta^{(i)} + \eta^{(i)} \hat{\otimes} \nabla^{(i)} + \frac{1}{2} \nabla \phi^{(i)} \hat{\otimes} \nabla \phi^{(i)} \right)_{AB} - \frac{1}{2} (\operatorname{tr} \chi)^{(i)} \widehat{\underline{\chi}}_{AB}^{(i)} \right), \\ \Omega \widehat{\chi}^{(i+1/2)}_{AB}(-1, v) = \frac{1}{2} \left(\Omega \widehat{\chi}_{AC} g^{CD} g^{(i)}_{DB} + \Omega \widehat{\chi}_{BC} g^{CD} g^{(i)}_{DA} \right) (-1, v). \end{cases} \quad (3.2)$$

Using (2.21), we define $\partial_v \phi^{(i+1)}$ by

$$\begin{cases} (\Omega e_3)^{(i)}(\partial_v \phi^{(i+1)}) + \frac{1}{2}(\Omega \operatorname{tr} \underline{\chi})^{(i)}(\partial_v \phi^{(i+1)}) \\ = (\Omega^{(i)})^2 \Delta^{(i)} \phi^{(i)} - \frac{1}{2}(\Omega \operatorname{tr} \chi)^{(i)}(\Omega e_3)^{(i)} \phi^{(i)} + 2(\Omega^{(i)})^2 \eta^{(i)} \nabla \phi^{(i)}, \\ \partial_v \phi^{(i+1)}(-1, v) = \partial_v \phi(-1, v). \end{cases} \quad (3.3)$$

It follows that

$$\nabla_A \phi^{(i+1)}(u, v) = \nabla_A \phi(u, 0) + \partial_A \left(\int_0^v \partial_v \phi^{(i+1)} \right).$$

2. Construction of $(\Omega \underline{\omega})^{(i+1/2)}$

We use (2.6) to define $(\Omega \underline{\omega})^{(i+1/2)}$,

$$\begin{cases} 4\partial_v(\Omega \underline{\omega})^{(i+1/2)} = \Omega \widehat{\chi}_{AB}^{(i+1/2)} \Omega \widehat{\chi}_{CD}^{(i)} (g^{(i)})^{AC} (g^{(i)})^{BD} + \frac{1}{2}(\Omega \operatorname{tr} \chi)^{(i)} (\Omega \operatorname{tr} \underline{\chi})^{(i)} \\ - 4(\Omega^{(i)})^2 \eta^{(i)} \cdot \underline{\eta}^{(i)} + (\partial_u + b^{(i)})(\Omega \operatorname{tr} \chi)^{(i)} - 2(\Omega^{(i)})^2 \operatorname{div}^{(i)} \eta^{(i)} \\ + (\partial_v \phi)^{(i+1)}(\Omega e_3 \phi)^{(i)}, \\ (\Omega \underline{\omega})^{(i+1/2)}(u, 0) = \Omega \underline{\omega}(u, 0). \end{cases} \quad (3.4)$$

3. Construction of $\Omega^{(i+1)}$, $\nabla_A \log \Omega^{(i+1)}$, and $(\Omega \omega)^{(i+1)}$

With $(\Omega \underline{\omega})^{(i+1/2)}$, we can integrate it to obtain $\Omega^{(i+1)}$

$$\begin{cases} (\partial_u + b^{(i)}) \log \Omega^{(i+1)} = -2(\Omega \underline{\omega})^{(i+1/2)}, \\ \log \Omega^{(i+1)}(-1, v) = \log \Omega(-1, v), \end{cases} \quad (3.5)$$

$$\nabla_A \log \Omega^{(i+1)} = \partial_A \log \Omega^{(i+1)}. \quad (3.6)$$

We construct $(\Omega \omega)^{(i+1)}$ by integration instead of differentiation,

$$\begin{cases} 4(\partial_u + b^{(i)})(\Omega \omega)^{(i+1)} = \Omega \widehat{\chi}_{AB}^{(i+1/2)} \Omega \widehat{\chi}_{CD}^{(i)} (g^{(i)})^{AC} (g^{(i)})^{BD} + \frac{1}{2}(\Omega \operatorname{tr} \chi)^{(i)} (\Omega \operatorname{tr} \underline{\chi})^{(i)} \\ - 4(\Omega^{(i)})^2 \eta^{(i)} \cdot \underline{\eta}^{(i)} + (\partial_u + b^{(i)})(\Omega \operatorname{tr} \chi)^{(i)} - 2(\Omega^{(i)})^2 \operatorname{div}^{(i)} \eta^{(i)} \\ + (\partial_v \phi)^{(i+1)}(\Omega e_3 \phi)^{(i)} - 8(\Omega^{(i)})^2 (\zeta^{(i)})^A \nabla_A \log \Omega^{(i+1)}, \\ (\Omega \omega)^{(i+1)}(-1, v) = \Omega \omega(-1, v). \end{cases} \quad (3.7)$$

We note that the construction of $(\Omega \omega)^{(i+1)}$ is equivalent to $(\Omega \omega)^{(i+1)} = -\frac{1}{2} \partial_v \log \Omega^{(i+1)}$, because

$$\begin{aligned} 4(\Omega e_3)^{(i)}(\Omega \omega)^{(i+1)} &= -2\partial_v(\Omega e_3)^{(i)} \log \Omega^{(i+1)} + 2\mathcal{L}_v b^{(i)} \nabla \log \Omega^{(i+1)} \\ &= 4\partial_v(\Omega \underline{\omega})^{(i+1/2)} - 8(\Omega^{(i)})^2 (\zeta^{(i)})^A \nabla_A \log \Omega^{(i+1)}. \end{aligned} \quad (3.8)$$

4. Construction of $\zeta^{(i+1)}$, $b^{(i+1)}$, and $(\Omega \underline{\omega})^{(i+1)}$

From (2.7), we define $\zeta^{(i+1)}$,

$$\begin{cases} \mathcal{L}_v(\zeta^{(i+1)})^A = -2(\Omega \operatorname{tr} \chi)^{(i)} (\zeta^{(i)})^A - 2(g^{(i)})^{AC} \left((\Omega \widehat{\chi})^{(i+1/2)} \right)_{CB} (\zeta^{(i)})^B \\ + 2(g^{(i)})^{AB} \partial_B (\Omega \omega)^{(i+1)} + (\operatorname{div})^{(i)} (\Omega \widehat{\chi})_B^{(i+1/2)} (g^{(i)})^{BA} \\ - \frac{1}{2} (g^{(i)})^{AB} \partial_B (\Omega \operatorname{tr} \chi)^{(i)} + (\Omega \operatorname{tr} \chi)^{(i)} (g^{(i)})^{AB} \partial_B \log \Omega^{(i+1)} \\ - (\partial_v \phi)^{(i+1)} \partial_B \phi^{(i)} (g^{(i)})^{AB}, \\ (\zeta^{(i+1)})^A(u, 0) = \zeta^A(u, 0). \end{cases} \quad (3.9)$$

We remark that it is necessary to treat $\zeta^{(i+1)}$ as a vector field instead of a 1-form in order to define $b^{(i+1)}$,

$$\begin{cases} \mathcal{L}_v(b^{(i+1)})^A = -4(\Omega^{(i+1)})^2(\zeta^{(i+1)})^A, \\ (b^{(i+1)})^A(u, 0) = b^A(u, 0). \end{cases} \quad (3.10)$$

5. Construction of $g^{(i+1)}$ and $(\Omega\chi)^{(i+1)}$

It remains to define $g^{(i+1)}$. We consider the following $(g, \text{tr } \chi, \widehat{\chi})$ system,

$$\begin{cases} \mathcal{L}_v(g^{(i+1)})_{AB} = (\Omega \text{tr } \chi)^{(i+1)}(g^{(i+1)})_{AB} + 2(\Omega\widehat{\chi})^{(i+1)}_{AB}, \\ \partial_v(\Omega \text{tr } \chi)^{(i+1)} = -\frac{1}{2}\left((\Omega \text{tr } \chi)^{(i+1)}\right)^2 - 4(\Omega\omega)^{(i+1)}(\Omega \text{tr } \chi)^{(i+1)} \\ \quad - (\Omega\widehat{\chi})^{(i+1)}_{AB}(\Omega\widehat{\chi})^{(i+1)}_{CD}g^{(i+1)AC}g^{(i+1)BD} - (\partial_v\phi^{(i+1)})^2, \\ (\Omega\widehat{\chi})^{(i+1)}_{AB} = \frac{1}{2}\left((\Omega\widehat{\chi})^{(i+1/2)}_{AC}(g^{(i)})^{CD}g^{(i+1)}_{DB} + (\Omega\widehat{\chi})^{(i+1/2)}_{BC}(g^{(i)})^{CD}g^{(i+1)}_{DA}\right), \\ (g^{(i+1)})_{AB}(u, 0) = g_{AB}(u, 0), \quad (\Omega \text{tr } \chi)^{(i+1)}(u, 0) = \Omega \text{tr } \chi(u, 0). \end{cases} \quad (3.11)$$

6. Construction of $(\Omega\chi)^{(i+1)}$ and $(\Omega e_3\phi)^{(i+1)}$

There are two equivalent ways to define $(\Omega\chi)^{(i+1)}$. The first one is using $\Omega\chi_{AB} = \frac{1}{2}\mathcal{L}_{\Omega e_3}g_{AB}$. We use the second way, which is using equation

$$\begin{cases} \mathcal{L}_v(\Omega\chi)^{(i+1)}_{AB} = \mathcal{L}_{\partial_u+b^{(i+1)}}(\Omega\chi)^{(i+1)}_{AB} - 2(\Omega^{(i+1)})^2\left(\nabla_A^{(i+1)}\zeta_B^{(i+1)} + \nabla_B^{(i+1)}\zeta_A^{(i+1)}\right) \\ \quad - 4(\Omega^{(i+1)})^2\left(\zeta_A^{(i+1)}\nabla_B \log \Omega^{(i+1)} + \zeta_B^{(i+1)}\nabla_A \log \Omega^{(i+1)}\right), \\ (\Omega\chi)^{(i+1)}_{AB}(u, 0) = \Omega\chi_{AB}(u, 0). \end{cases} \quad (3.12)$$

We remark that

$$\begin{aligned} & \Omega\nabla_4(\Omega\chi)_{AB} - \Omega\nabla_3(\Omega\chi)_{AB} \\ &= \mathcal{L}_v(\Omega\chi)_{AB} - (\Omega\chi)_A^C(\Omega\chi)_{CB} - (\Omega\chi)_B^C(\Omega\chi)_{CA} \\ & \quad - \mathcal{L}_{\partial_u+b}(\Omega\chi)_{AB} + (\Omega\chi)_A^C(\Omega\chi)_{CB} + (\Omega\chi)_B^C(\Omega\chi)_{AC} \\ &= \mathcal{L}_v(\Omega\chi)_{AB} - \mathcal{L}_{\partial_u+b}(\Omega\chi)_{AB}. \end{aligned} \quad (3.13)$$

Theoretically, $(\Omega\chi)^{(i+1)}_{AB} = \frac{1}{2}\mathcal{L}_{\partial_u+b^{(i+1)}}g^{(i+1)}_{AB}$. Similarly, we construct $(\Omega e_3\phi)^{(i+1)}$ by

$$\begin{cases} \partial_v(\Omega e_3\phi)^{(i+1)} = (\partial_u + b^{(i+1)})\partial_v\phi^{(i+1)} - 4(\Omega^{(i+1)})^2(\zeta^{(i+1)})^A\nabla_A\phi^{(i+1)}, \\ (\Omega e_3\phi)^{(i+1)}(u, 0) = \Omega e_3\phi(u, 0). \end{cases} \quad (3.14)$$

For Einstein vacuum equation

$$\text{Ric}_{\mu\nu} = 0,$$

removing all ϕ -terms of ESE gives the iterative construction directly, so we omit the repetition of construction.

3.2 Curvature residues

Computing curvature usually involves second order derivatives, which would heavily reduce the numerical precision. Especially, $\lim_{v \rightarrow 0} R_{4A4B}(-1, v)$ can be infinite for short pulse perturbation and Ric_{44} is hard to compute in this way, even though it is theoretically zero in our construction. See [1] for the example of short pulse with infinite R_{4A4B} . We use intrinsic equations of Lorentzian manifold to compute the Ricci curvature:

$$\Omega^2 \text{Ric}_{33} = -\Omega\nabla_3(\Omega \text{tr } \chi) - 4\Omega\omega\Omega \text{tr } \chi - \frac{1}{2}(\Omega \text{tr } \chi)^2 - |\Omega\widehat{\chi}|^2,$$

$$\begin{aligned}
\Omega \operatorname{Ric}_{3A} &= \Omega \nabla_3 \zeta + \frac{3}{2} \Omega \operatorname{tr} \underline{\chi} \zeta + \Omega \widehat{\underline{\chi}} \cdot \zeta + 2 \nabla(\Omega \underline{\omega}) + \operatorname{div}(\Omega \widehat{\underline{\chi}}) - \frac{1}{2} \nabla(\Omega \operatorname{tr} \underline{\chi}) + \Omega \operatorname{tr} \underline{\chi} \nabla \log \Omega, \\
\Omega \operatorname{Ric}_{4A} &= -\Omega \nabla_4 \zeta - \frac{3}{2} \Omega \operatorname{tr} \chi \zeta - \Omega \widehat{\chi} \cdot \zeta + 2 \nabla(\Omega \omega) + \operatorname{div}(\Omega \widehat{\chi}) - \frac{1}{2} \nabla(\Omega \operatorname{tr} \chi) + \Omega \operatorname{tr} \chi \nabla \log \Omega, \\
\Omega^2 \widehat{\operatorname{Ric}}_{AB} &= \Omega \nabla_3(\Omega \widehat{\chi}) + \frac{1}{2} \Omega \operatorname{tr} \underline{\chi} \Omega \widehat{\chi} - \Omega^2 (\nabla \hat{\otimes} \eta + \eta \hat{\otimes} \eta) + \frac{1}{2} \Omega \operatorname{tr} \chi \Omega \widehat{\underline{\chi}}, \\
\Omega^2 g^{AB} \operatorname{Ric}_{AB} &= \Omega \nabla_3(\Omega \operatorname{tr} \chi) + \Omega \operatorname{tr} \chi \Omega \operatorname{tr} \underline{\chi} - 2 \Omega^2 \operatorname{div} \eta - 2 \Omega^2 |\eta|^2 + 2 \Omega^2 K, \\
\Omega^2 \operatorname{Ric}_{34} &= 4 \Omega \nabla_4(\Omega \underline{\omega}) - \Omega \widehat{\chi} \cdot \Omega \widehat{\underline{\chi}} - \frac{1}{2} \Omega \operatorname{tr} \chi \Omega \operatorname{tr} \underline{\chi} + 4 \Omega^2 \eta \cdot \underline{\eta} - \Omega \nabla_3(\Omega \operatorname{tr} \chi) + 2 \Omega^2 \operatorname{div} \eta.
\end{aligned}$$

We only need the first order derivatives of the iterated quantities in this way.

3.3 Legendre–Gauss–Lobatto spectral elements

In the experiments, we will add non-spherical symmetric perturbation to the Einstein scalar-field or vacuum systems. To make the metric less regular, the perturbation is required to be of size $O(v^\delta)$. To deal with the non-smoothness, the characteristic coordinates are first replaced by coordinates in which the corner profiles are smoother. In particular, we use

$$\tau = -\log(-u), \quad s = \left(\frac{v}{v_{\max}} \right)^\delta, \quad u = -e^{-\tau}, \quad v = v_{\max} s^{1/\delta}.$$

Thus a profile proportional to v^δ is linear in s . This change of variables does not regularize the physical solution; it only represents its fractional behavior by a smooth function of the computational coordinate. Physical derivatives and integrals are recovered by

$$\partial_u = \frac{1}{-u} \partial_\tau, \quad \partial_v = \left(\frac{dv}{ds} \right)^{-1} \partial_s, \quad \int_0^v F(v') dv' = \int_0^s F(v(s')) \frac{dv}{ds'} ds'.$$

We divide each computational coordinate interval into elements. On an element of degree p , let $\{\xi_j\}_{j=0}^p$ be the Legendre–Gauss–Lobatto nodes, characterized by

$$(1 - \xi_j^2) P'_p(\xi_j) = 0,$$

including the two endpoints. A component F is represented by its Lagrange interpolant

$$I_p F(\xi) = \sum_{j=0}^p F(\xi_j) \ell_j(\xi).$$

The element differentiation and indefinite-integration matrices are

$$D_{ij} = \ell'_j(\xi_i), \quad Q_{ij} = \int_{-1}^{\xi_i} \ell_j(\xi) d\xi.$$

Consequently, interpolation of intermediate Runge–Kutta stages, differentiation, and cumulative integration are all obtained from the same element polynomial. Neighboring elements share their endpoint value, while the two derivative traces at an interior interface are combined into a single interface derivative. This composite construction permits local refinement in either characteristic direction without replacing the underlying transport hierarchy.

3.4 Tensor spherical-harmonic Galerkin discretization

The angular variables are represented on a quasi-uniform, pole-free point set on $\mathbb{S}^2$. Geometric vectors and tensors are stored as ambient Cartesian tensors and projected to the tangent bundle. This avoids the coordinate singularities of a polar chart. For a scalar field, we use the real spherical-harmonic expansion

$$f_L(\vartheta) = \sum_{\ell=0}^L \sum_{m=-\ell}^{\ell} f_{\ell m} Y_{\ell m}(\vartheta).$$

The coefficients are obtained by an overdetermined least-squares analysis of the nodal values, and angular derivatives are computed by differentiating the harmonic basis.

Vectors and symmetric two-tensors are not expanded componentwise as unrelated scalars. We use the electric and magnetic vector harmonics

$$Y_A^{E,\ell m} \sim \nabla_A Y_{\ell m}, \quad Y_A^{B,\ell m} \sim \epsilon_A^B \nabla_B Y_{\ell m},$$

and the trace, electric, and magnetic tensor harmonics

$$\gamma_{AB} Y_{\ell m}, \quad Y_{AB}^{E,\ell m} \sim (\nabla_A \nabla_B Y_{\ell m})^{\text{TF}}, \quad Y_{AB}^{B,\ell m} \sim \epsilon_{(A}^C \nabla_{B)} \nabla_C Y_{\ell m}.$$

These typed spaces preserve tangency, symmetry, and the tensorial transformation law. Trace-free quantities are projected with respect to the current section metric,

$$T_{AB}^{\text{TF}} = T_{AB} - \frac{1}{2}(\text{tr}_\gamma T)\gamma_{AB},$$

rather than with respect to a fixed background metric.

Let $\mathcal{V}_L$ denote the retained harmonic space and let $\mathcal{V}_W$, with $W > L$, be a work space used to form angular products. For a semidiscrete unknown $U_L \in \mathcal{V}_L$, every nonlinear right-hand side is defined by

$$\partial U_L = \Pi_L F(U_L),$$

where $F(U_L)$ is evaluated in the work space and Π_L is the appropriate scalar, vector, or tensor Galerkin projection. The retained/work separation reduces angular aliasing while keeping the evolved state in one declared finite-dimensional geometric space.

4 Results for Einstein Vacuum Equations

5 Results for Einstein Scalar-field Equations

6 Conclusion

References

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